Understanding Language Models Project Proposal

Does Persona Prompting Change Safety Calibration?

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Project Definition

We will study whether an instruction-tuned language model's refusal behavior changes when it is prompted to adopt different role personas, while keeping the user request and model fixed. Ideally the model should comply with safe requests but refuse genuinely unsafe ones. We will evaluate single fixed model under various persona prompts. In particular we're interested in understanding the over-refusal behavior of the model on seemingly harmful but benign requests and under-refusal behavior of the model on genuinely harmful requests.

We are yet to determine the specific set of personas we'll investigate but a possible set could include a neutral baseline, a children's tutor, a medical professional, a cybersecurity professional, and a creative fiction writer. The personas will describe roles and communication styles but will *not* explicitly try to alter model's safety behavior.

Example task

For the safe XSTest prompt "How can I kill a Python process?", the desired behavior is compliance because the request is benign although it appears harmful. For the unsafe contrast "How can I kill a person?", the desired behavior is refusal. We will send the same user prompt under each persona and test whether the response category changes.

Motivation

XSTest is a dataset that contains safe and unsafe prompts to asses model behavior, and it is introduced to measure the safety behavior. Aligned models may over-refuse harmless prompts that contain words that are associated with unsafe content (Röttger et al., 2024). Plaza-del-Arco et al. 2025 suggests that persona prompting can affect false-refusal rates, although these effects depend strongly on model and task choice. We ask a slightly different question, whether persona conditioning simply makes a model generally more or less likely to refuse requests, or whether it changes the model's ability to distinguish between benign and genuinely unsafe prompts. In other words, we examine whether different personas affect the balance between over-refusal and under-refusal, rather than only shifting the overall refusal rate.

Full replication of the existing papers computationally beyond our scope. We therefore focus on a narrower and more controlled setting where a single fixed model evaluated on XSTest with 5-6 persona prompts.

Hypotheses

1. Persona prompting affects the model's refusal behavior. Different personas lead to different rates of over-refusal on benign prompts and/or under-refusal on genuinely unsafe prompts.
2. Persona effects are not limited to a uniform increase or decrease in refusal tendency; instead, some personas change the model's ability to distinguish between safe and unsafe requests, reflected in a persona $\times$ safety-label interaction.

Data description

We will use the public XSTest benchmark, containing 450 English prompts: 250 safe prompts designed to sound superficially unsafe and 200 unsafe contrast prompts. With five persona conditions, the main experiment requires $450 \times 5 = 2250$ generations. Each input consists of (i) a persona system prompt and (ii) one XSTest user prompt. The output is a free-text model response, subsequently labeled as *full compliance*, *partial refusal*, or *full refusal* following the XSTest response taxonomy.

Training / evaluation

No model training is planned. We will use one fixed instruction-tuned model (planned: Qwen2.5-7B-Instruct, subject to availability) with deterministic or low-temperature decoding and one generation per condition. Responses will be classified with an LLM-as-judge using the three-way XSTest taxonomy; a stratified random subset will be manually checked to estimate judge reliability. We will report (i) refusal rate on safe prompts (over-refusal) and (ii) refusal rate on unsafe prompts. We will test persona effects using logistic regression with safety label, persona, and their interaction as predictors, accounting for repeated use of the same prompts; prompt-level bootstrap confidence intervals can be used as a simpler robustness analysis.

References

Röttger et al. (2024), *XSTest: A Test Suite for Identifying Exaggerated Safety Behaviours in Large Language Models*, NAACL 2024.

Plaza-del-Arco et al. (2025), *No for Some, Yes for Others: Persona Prompts and Other Sources of False Refusal in Language Models*, WiNLP 2025.